The Media Connectivity Association




OVERVIEW AND UPDATES · CONTENT DELIVERY AND STREAMING

# Video Operation Points

Video capabilities and operation points of 3GPP TS 26.265, and the conformance validator for them

3GPP

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# About the project

What 3GPP TS 26.265 defines, and the part the validator comes from

## Video capabilities

**Part of 3GPP TS 26.265**

Service-independent video decoding and encoding capabilities, which 3GPP service specifications and third-party services may reference.

## Operation points

**Part of 3GPP TS 26.265**

Combinations of a video coding format and a representation format, with requirements for the bitstream and for the receiver.

## Conformance

**Part of 3GPP TS 26.265**

The conformance functions, among them the conformance validator that tests whether a bitstream conforms to an operation point.

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# Operation points

The seven operation points of TS 26.265, grouped by video format

### 3GPP-HD

#### AVC and HEVC

3GPP-AVC-HD: AVC High Progressive Profile, Level 4.0. 3GPP-HEVC-HD: HEVC Main 10 Profile, Main Tier, Level 4.1.

### 3GPP-HDR

#### HEVC, HD and UHD

HEVC Main 10 Profile, Main Tier: 3GPP-HEVC-HD-HDR at Level 4.1 and 3GPP-HEVC-UHD-HDR at Level 5.1.

### 3GPP-STEREO

#### HEVC and MV-HEVC

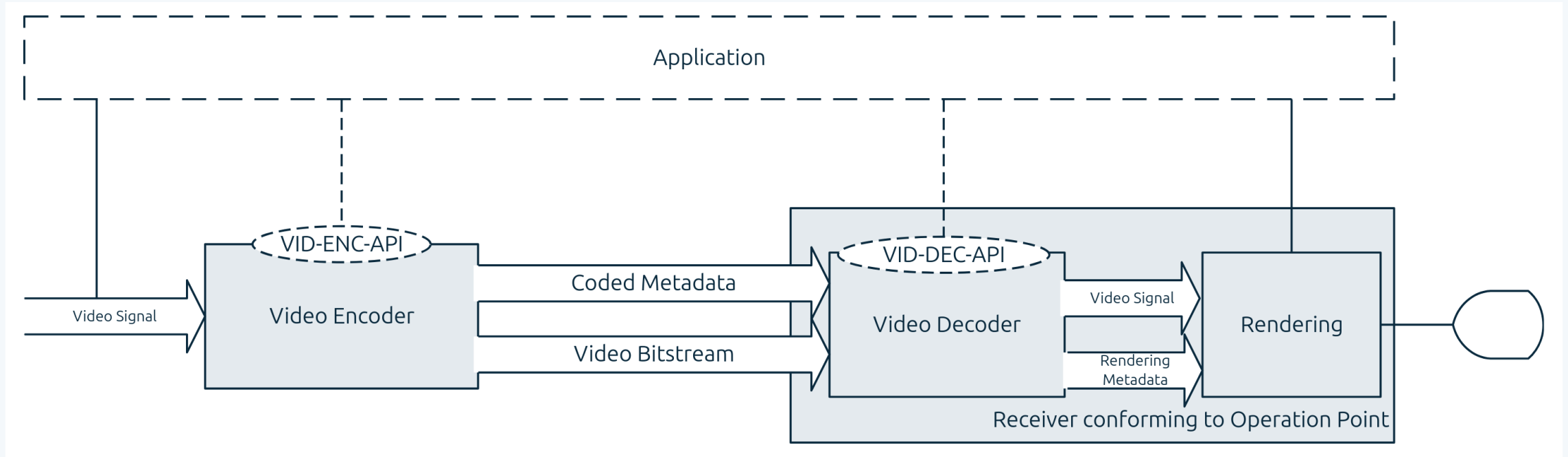
3GPP-HEVC-Stereo: HEVC Main 10 Profile, Main Tier, Level 5.2, frame packed. 3GPP-MV-HEVC-Main-Stereo and 3GPP-MV-HEVC-Ext-Stereo: MV-HEVC, two layers.

**An operation point is a collection of discrete combinations of video representation formats: spatial and temporal resolutions, colour mapping, transfer functions and the encoding format.**

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# Reference architecture

From a video signal to rendering, in the specification




Redrawn from TS 26.265. The APIs are not normatively specified: they serve as an example reference to configure encoders and decoders.

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# Conformance functions

What the conformance part of TS 26.265 provides

| Test material | Validator | Conforming decoders | Conformance bitstreams |
| --- | --- | --- | --- |
| Annotated video test signals, and reference encoders with configurations. | Tests bitstreams that claim to conform, and reports which test assertions are not fulfilled. | Return a report and an MD5. | Give a decoder under test a bitstream, and the decoder conformance test its information. |

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# How the validator works

It parses a bitstream to XML, then checks the XML against schemas

| 1 Parse bitstream | 2 Generate XML | 3 Load XSD schemas | 4 Validate |
|---|---|---|---|
| Reads the NAL units of the AVC or HEVC bitstream and parses their headers and parameters. | Creates an XML element for each NAL unit, with its parameters nested by codec structure. | Reads the operation point and loads the XSD 1.1 schemas specified for it. | Runs the XSD validation, extracts which assertions pass or fail and writes the report. |

From the repository README. A "shall" requirement that fails is reported as an error, a "should" as a warning.

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# How it grew

Video operation points before and in TS 26.265, release by release

## Release 15 to 18

Earlier operation points in other specifications: HDR and VR profiles (TS 26.116, TS 26.118), an 8K TV operation point (TS 26.116, TS 26.511).

## Release 19

TS 26.265 is created by the VOPS work item: operation points and capabilities consolidated, with MV-HEVC stereo operation points and conformance assets.

## Release 20

Release 20 (work plan of 28 September 2026): no work item lists TS 26.265. The study FS_AVFOPS_MED (TR 26.966) is listed; its relation to TS 26.265 is not stated.

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# What is still open

Points the specification leaves for further study, which bound the validator

## System integration

**Definitions for AVC and HEVC**

The specification states that the definitions for AVC and for HEVC are for further study.

## Encoder API

**Mapping of operation points**

The mapping of operation points to an encoder API is stated as work for further study.

## Validator report

**Semantics and format**

The detailed semantics and the format of the report are for further study, so the JSON report is the validator's own.

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# Repositories

The conformance validator, from the description of its repository

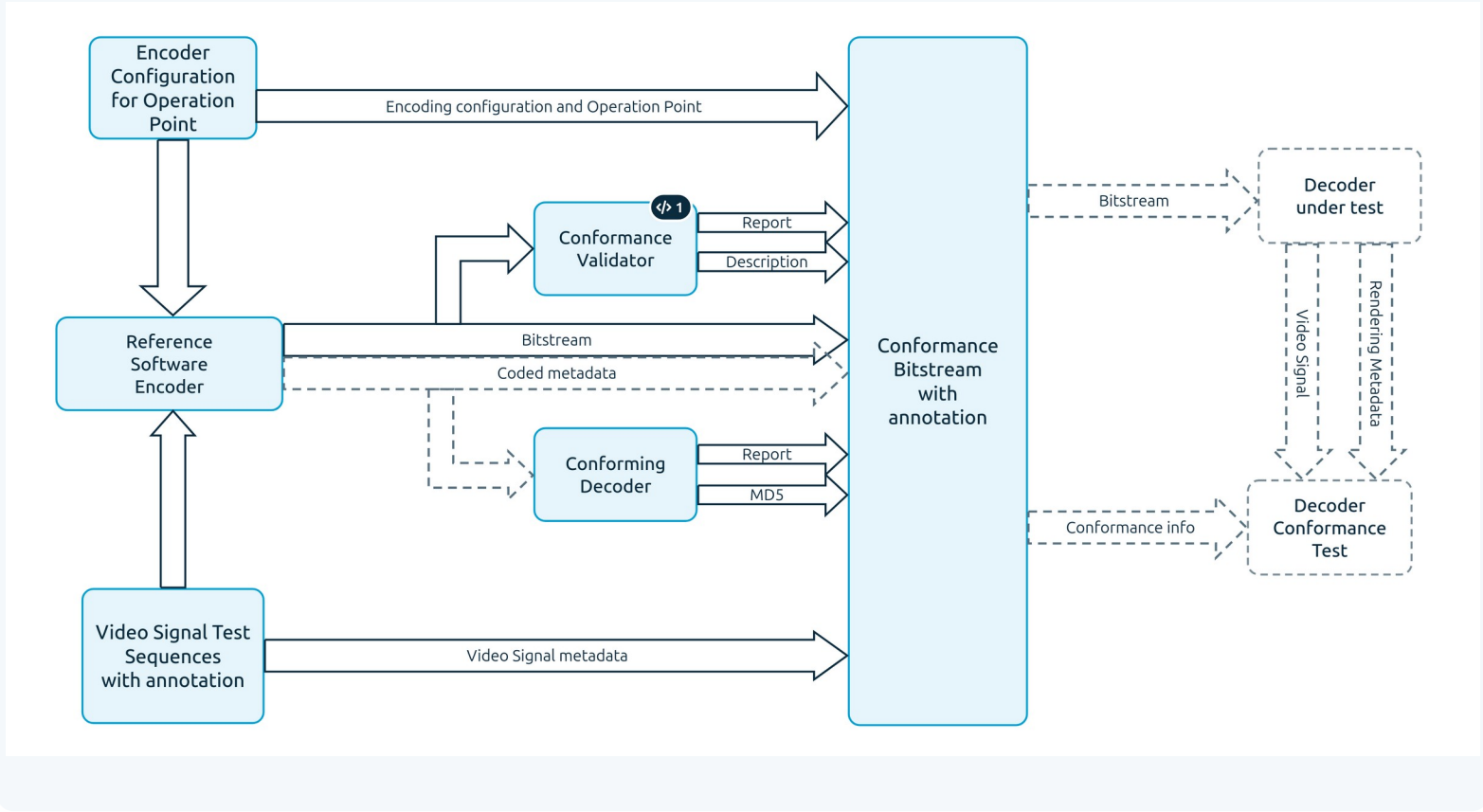
| REPOSITORY | WHAT IT IS | COVERS | LATEST RELEASE | |
|---|---|---|---|---|
| vops-conformance-validator | Checks AVC, HEVC and MV-HEVC bitstreams against the operation points of TS 26.265; some requirements only partly | AVC, HEVC, MV-HEVC, XSD 1.1 schemas, JSON report | none yet | GitHub |

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released no release yet

# Repositories in the architecture

The Reference Tools on the specification figure, as the figure generator draws them




## PROVIDED BY THE REFERENCE TOOLS

1 **vops-conformance-validator**

Only the Conformance Validator is mapped to a repository.

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# Latest updates and roadmap

Status on 22 October 2026

## PLANNED

### Testing with conformance bitstreams

Every operation point is to be tested with its conformance bitstreams, to confirm the schemas and parsers.

### Version 19.3.0 of the specification

The Standards page keeps V19.2.0 as the baseline until the validator is rebuilt against V19.3.0, which changes the dual-layer HEVC capabilities.

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